

Physics-0 2026 Spring

Homework 7

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*All of my homework is available at <https://github.com/lacampanella233/Homework.git>.

Problem 1. (30pts) Consider a spherically symmetric planet of radius R and uniform mass density ρ .

(a) (20pts) Find the potential energy of a particle of mass m at a position $\mathbf{r}$ in the interior of the planet.

(b) (10pts) Compute the force associated with this potential. The force should agree with the expression computed in class using Gauss' law.

Solution. (a) By the spherical symmetry of the problem, the potential energy of the particle at a position $\mathbf{r}$ in the interior of the planet only depends on the distance r from the center of the planet. Then the potential energy can be written as $\varphi(\mathbf{r}) = \varphi(r)$.

φ can be solved by the Poisson's equation in the spherical coordinates:

$$\nabla^2 \varphi(\mathbf{r}) = 4\pi G \rho \quad (r < R) \implies \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\varphi(r)}{dr} \right) = 4\pi G \rho \implies \varphi(r) = \frac{2}{3} \pi G \rho r^2 + C.$$

We will take $C = 0$. Then the potential energy of the particle is

$$m\varphi(\mathbf{r}) = \frac{2}{3} \pi G m \rho r^2.$$

(b) The force can be computed by

$$\mathbf{F} = -\nabla \varphi(\mathbf{r}) = -\frac{d\varphi(r)}{dr} \mathbf{e}_r = -\frac{4}{3} \pi G \rho r \mathbf{e}_r.$$

Problem 2. (20pts) Consider a uniform, i.e. constant, electric field $\mathbf{E}$.

(a) (10pts) Find the potential energy of a particle with charge q in this electric field.

(b) (10pts) Using the expression above, find the potential energy of a dipole with moment $\mathbf{p}$ in this electric field.

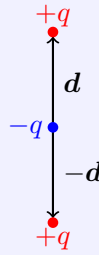
Solution. (a) Select the coordinate system such that the electric field is along the x -axis, i.e. $\mathbf{E} = E \mathbf{e}_x$. Then the potential energy of a particle with charge q at $\mathbf{r} = (x, y, z)$ is given by

$$\mathcal{E} = -q \mathbf{E} \cdot \mathbf{r} = -q E x.$$

(b) An electric dipole with moment $\mathbf{p}$ can be considered as two opposite charges $+q$ and $-q$ separated by a small distance $\mathbf{d}$, such that $\mathbf{p} = q \mathbf{d}$. The potential energy of the dipole in the electric field can be computed by

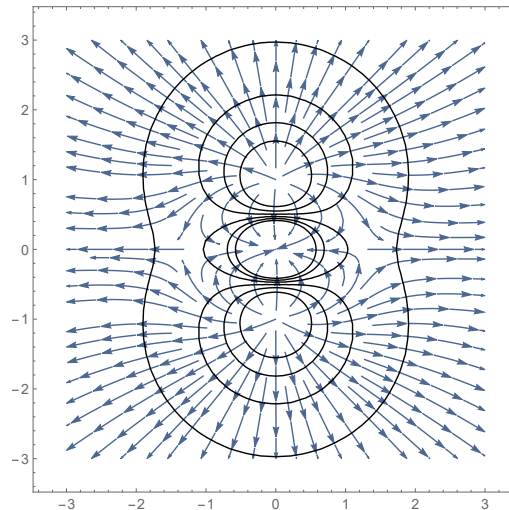
$$\mathcal{E} = -q(\mathbf{r} + \mathbf{d}) \cdot \mathbf{E} - (-q) \mathbf{r} \cdot \mathbf{E} = -\mathbf{p} \cdot \mathbf{E}.$$

Problem 3. (50pts) Consider the configuration of charges shown in the diagram below where q is a positive number.



- (a) (10pts) Sketch the electric field lines.
- (b) (10pts) Write the electric field $\mathbf{E}$ as a function of the radial coordinate $\mathbf{r}$ (make sure to indicate the origin of your coordinate system).
- (c) (15pts) The electric field $\mathbf{E}$ at a distance $r \gg |d|$ can be expanded in powers of $1/r$. The leading term in the expansion falls off as $1/r^n$ for some integer n . Find the leading term. Explain why we should expect this result.
- (d) (15pts) Compute the next term in the expansion of the electric field at a distance $r \gg |d|$, that is, the term that goes like $1/r^{n+1}$, where n was determined above. Explain why we should expect this result.

Solution. (a) The diagram below shows the electric field lines and the isolines of the electric potential.



(b) We use the coordinate system with the origin at the position of the negative charge, and the positive charges are located at $\mathbf{d}$ and $-\mathbf{d}$. The electric field at a point $\mathbf{r}$ can be computed by superposition:

$$\mathbf{E}(\mathbf{r}) = q \frac{\mathbf{r} - \mathbf{d}}{|\mathbf{r} - \mathbf{d}|^3} + q \frac{\mathbf{r} + \mathbf{d}}{|\mathbf{r} + \mathbf{d}|^3} - q \frac{\mathbf{r}}{|\mathbf{r}|^3}.$$

(c) Denote $\hat{\mathbf{r}} = \mathbf{r} / |\mathbf{r}|$ and $r = |\mathbf{r}|$. Then

$$\begin{aligned} \mathbf{E}(\mathbf{r}) &= \frac{1}{r^2} \left(q \frac{\hat{\mathbf{r}} - \mathbf{d}/r}{|\hat{\mathbf{r}} - \mathbf{d}/r|^3} + q \frac{\hat{\mathbf{r}} + \mathbf{d}/r}{|\hat{\mathbf{r}} + \mathbf{d}/r|^3} - q \frac{\hat{\mathbf{r}}}{|\hat{\mathbf{r}}|^3} \right) \\ &\approx \frac{q}{r^2} \left\{ \left(\hat{\mathbf{r}} - \frac{1}{r} \mathbf{d} \right) \left(1 + 3\hat{\mathbf{r}} \cdot \frac{\mathbf{d}}{r} \right) + \left(\hat{\mathbf{r}} + \frac{1}{r} \mathbf{d} \right) \left(1 - 3\hat{\mathbf{r}} \cdot \frac{\mathbf{d}}{r} \right) - \hat{\mathbf{r}} \right\} \\ &\approx \frac{q}{r^2} \left(\hat{\mathbf{r}} - \frac{6\hat{\mathbf{r}} \cdot \mathbf{d}}{r^2} \mathbf{d} \right). \end{aligned}$$

The leading term is

$$\mathbf{E}^{(0)}(\mathbf{r}) = \frac{q}{r^2} \hat{\mathbf{r}}.$$

This is because the total charge of the system is $+q$.

(d) The next term in the expansion is

$$\mathbf{E}^{(1)}(\mathbf{r}) = 0.$$

This is because the dipole moment of the system is zero.